Identity needed to prove MLS Theorem 2.17 p. 65

To prove theorem 2.17, the following identity is used, where $\vec{\omega}$ is a unit vector:

$$(\vec{\omega} \times \vec{\tau}) \times \vec{\omega} + (\vec{\omega} \cdot \vec{\tau})\vec{\omega} = \vec{\tau}$$

or, if $\vec{\omega}$ and $\vec{\tau}$ are written in component form as the two 3×1column vectors ω and τ, respectively,

$$\widehat{(\widehat{\omega}\tau)}\ \omega + (\omega^T\tau)\omega = \tau\ .$$

This code proves this identity.

```
clearvars, clc

syms omega1 omega2 omega3 real
syms tau1 tau2 tau3 real

omega   = [omega1; omega2; omega3]
```

omega =

$$\begin{pmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{pmatrix}$$

```
tau     = [tau1; tau2; tau3]
```

tau =

$$\begin{pmatrix} \tau_1 \\ \tau_2 \\ \tau_3 \end{pmatrix}$$

```
h_omega = hatm_sym(omega) % Function to creates skew-symmetric matric of
```

h_omega =

$$\begin{pmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{pmatrix}$$

```
                          %    argument.

dum1    = h_omega*tau
```

dum1 =

$$\begin{pmatrix} \omega_2\,\tau_3 - \omega_3\,\tau_2 \\ \omega_3\,\tau_1 - \omega_1\,\tau_3 \\ \omega_1\,\tau_2 - \omega_2\,\tau_1 \end{pmatrix}$$

```
term1   = expand(hatm_sym(dum1)*omega)
```

term1 =

$$\begin{pmatrix} \tau_1\,{\omega_2}^2 - \omega_1\,\tau_2\,\omega_2 + \tau_1\,{\omega_3}^2 - \omega_1\,\tau_3\,\omega_3 \\ \tau_2\,{\omega_1}^2 - \omega_2\,\tau_1\,\omega_1 + \tau_2\,{\omega_3}^2 - \omega_2\,\tau_3\,\omega_3 \\ \tau_3\,{\omega_1}^2 - \omega_3\,\tau_1\,\omega_1 + \tau_3\,{\omega_2}^2 - \omega_3\,\tau_2\,\omega_2 \end{pmatrix}$$

```
term2   = expand((omega'*tau)*omega)
```

term2 =

$$\begin{pmatrix} {\omega_1}^2\,\tau_1 + \omega_1\,\omega_2\,\tau_2 + \omega_1\,\omega_3\,\tau_3 \\ {\omega_2}^2\,\tau_2 + \omega_1\,\omega_2\,\tau_1 + \omega_2\,\omega_3\,\tau_3 \\ {\omega_3}^2\,\tau_3 + \omega_1\,\omega_3\,\tau_1 + \omega_2\,\omega_3\,\tau_2 \end{pmatrix}$$

```
sum1 = term1 + term2
```

sum1 =

$$\begin{pmatrix} \tau_1\,{\omega_1}^2 + \tau_1\,{\omega_2}^2 + \tau_1\,{\omega_3}^2 \\ \tau_2\,{\omega_1}^2 + \tau_2\,{\omega_2}^2 + \tau_2\,{\omega_3}^2 \\ \tau_3\,{\omega_1}^2 + \tau_3\,{\omega_2}^2 + \tau_3\,{\omega_3}^2 \end{pmatrix}$$

```
factors1 = subs(factor(sum1(1,1)),omega'*omega,1)
```

factors1 = $\begin{pmatrix} \tau_1 & 1 \end{pmatrix}$

```
factors2 = subs(factor(sum1(2,1)),omega'*omega,1)
```

factors2 = $\begin{pmatrix} \tau_2 & 1 \end{pmatrix}$

```
factors3 = subs(factor(sum1(3,1)),omega'*omega,1)
```

factors3 = $\begin{pmatrix} \tau_3 & 1 \end{pmatrix}$

```
sum1 = [prod(factors1);prod(factors2);prod(factors3)]
```

sum1 =

$$\begin{pmatrix} \tau_1 \\ \tau_2 \\ \tau_3 \end{pmatrix}$$